

# A Decoder-Only Jacobian Lens for Speech-to-Text Models

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**Abstract**—The Jacobian lens identifies representations a language model is poised to verbalize by averaging causal maps from a layer’s residual stream to the final pre-logit state. Encoder-decoder speech recognizers split that residual across two stacks: an encoder that is never unembedded and a decoder that is. We show that a pooled encoder-to-decoder transport is not this object—it destroys the causal  $(t, t')$  structure and leaves the unembedding off-manifold. We instead fit one Jacobian per decoder block,  $J_\ell = \mathbb{E}_{t, t' \geq t} [\partial h_{\text{final}, t'} / \partial h_{\ell, t}]$ , with Hutchinson probes, and read out  $\text{softmax}(E J_\ell h_{\ell, t})$  through Whisper’s own tied projection. The construction is implemented in ECHO. On a linear stub decoder the estimator recovers the closed form  $J = 2M/(T + 1)$  to relative Frobenius error  $< 0.1$ ; a Whisper-base fit yields six  $512 \times 512$  maps. We specify a reading protocol for the resulting (position  $\times$  layer) grid and leave a workspace-band census as future work.

**Index Terms**—speech recognition, mechanistic interpretability, Jacobian lens, Whisper, encoder-decoder Transformers

## I. INTRODUCTION

Whisper [1] and related encoder-decoder ASR models transcribe by unembedding a decoder residual stream that has attended to a frozen acoustic encoder. Failures such as silence-induced repetition, homophone substitution, and early cutoff are well known, yet the decoder state that is about to become the next token is rarely inspected in vocabulary space.

Gurnee et al. [2] give language models a tool for that inspection: the Jacobian lens averages  $\partial h_{\text{final}, t'} / \partial h_{\ell, t}$  over causally allowed pairs  $t' \geq t$ , then reads the transported residual with the model’s unembedding. The ranked tokens are what the model is *poised to verbalize*. The construction assumes a single residual stream that ends in an unembedding. Speech-to-text models do not: the encoder never sees that unembedding.

This paper ports the estimator to the stream that does. We fit decoder-only Jacobians for seq2seq ASR, estimate them with Hutchinson probes [3] so the encoder can sit outside the autograd graph, and ship fit/apply as jobs in ECHO, an audio interpretability workbench in the spirit of LIT [4]. We also record why a pooled encoder transport—the obvious first attempt—is a different, weaker object. We do not claim that Whisper has a global workspace; we claim that if one wants the LLM lens, one must fit it on the decoder.

## II. RELATED WORK

The logit lens [5] applies the unembedding to intermediate residuals (identity transport). The tuned lens [6] learns an affine map to the final layer. The Jacobian lens [2] is the

average *derivative* of the true residual-to-final map. We copy that estimator, not the LLM write/hold/ablate suite.

DecoderLens [7] feeds intermediate *encoder* layers to a frozen decoder’s cross-attention so the decoder can verbalize encoder states. Langedijk et al. already run this on Whisper transcription: early encoder layers emit empty or repetitive strings; later ones approximate the transcript. Complementary: they interpret the encoder *through* the decoder; we interpret the decoder *with* its own Jacobian. Linear probes [8]–[10] test linearly available features but are not maps onto the model’s unembedding.

ECHO is the audio analogue of LIT [4]: attention, saliency, embeddings, and perturbations on Whisper and wav2vec 2.0 [11]. Attention says where the decoder looked in the audio; saliency says which waveform regions moved a scalar. Neither says what the decoder is poised to say. CTC models have no decoder residual to unembed and are excluded.

## III. METHOD

### A. Substrate

Whisper is an encoder-decoder Transformer [12]. A 30 s log-Mel spectrogram (80 bins  $\times$  3000 frames on the base model) is downsampled by a stride-2 convolution to  $n_{\text{audio}} = 1500$  encoder positions of width  $d = 512$ . The decoder, at each transcript position  $t \leq 448$ , attends causally to its prefix and, via cross-attention, to all 1500 encoder frames, then unembeds with a tied projection  $E \in \mathbb{R}^{V \times d}$  ( $V = 51865$ ). We write  $h_{\ell, t} \in \mathbb{R}^d$  for the residual at decoder block  $\ell$  and position  $t$ , and  $h_{\text{final}, t'}$  for the post-final-LayerNorm state that  $E$  consumes.

Current HuggingFace Whisper records decoder hidden states from each decoder block only. The embedding output is not in that tuple;  $h_{\text{final}}$  is a distinct tensor (final LayerNorm of the last block). The lens therefore has *one map per decoder block*: six on Whisper-base, matching a fitted artifact on this codebase. The encoder is computed once without gradients and is never differentiated.

### B. Estimator

Following [2], the lens at block  $\ell$  is

$$J_\ell = \mathbb{E}_{t, t' \geq t, \text{clip}} \left[ \frac{\partial h_{\text{final}, t'}}{\partial h_{\ell, t}} \right] \in \mathbb{R}^{d \times d}. \quad (1)$$

The model is frozen. There is no optimizer and no learned baseline. Fit is teacher-forced: labels are shifted right past the decoder-start token so all  $T$  positions run in one pass.

TABLE I  
POOLED ENCODER TRANSPORT VERSUS THE DECODER-ONLY LENS.

|         | Encoder transport             | Decoder-only J-lens              |
|---------|-------------------------------|----------------------------------|
| Source  | mean of 1500 encoder frames   | residual $h_{\ell,t}$            |
| Target  | mean of $T$ decoder positions | $h_{\text{final},t'}, t' \geq t$ |
| Spaces  | encoder $\rightarrow$ decoder | decoder $\rightarrow$ decoder    |
| Readout | $(h - \bar{h})J^\top E^\top$  | $\text{softmax}(EJh)$            |
| Units   | 0.3 s buckets                 | decoder tokens                   |
| CTC     | yes                           | no                               |

Exact Jacobians are unnecessary. For a Rademacher probe  $r \in \{\pm 1\}^d$ , the scalar  $\sum_{t'} \langle h_{\text{final},t'}, r \rangle$  has vector-Jacobian product  $g_t = \sum_{t' \geq t} J_\ell(t \rightarrow t')^\top r$  at every source  $t$  at once, because the causal mask zeroes  $t > t'$ . The identity  $\mathbb{E}[rr^\top] = I$  then gives

$$\hat{J}_\ell = \frac{1}{NP \cdot T(T+1)/2} \sum_{n=1}^N \sum_{p=1}^P r^{(p)} \left( \sum_t g_t^{(n,p)} \right)^\top, \quad (2)$$

with  $N$  clips and  $P$  probes [3]. Product defaults are  $P = 4$  (range 1–32) and  $N \in [2, 1000]$ . Audio is truncated at 30 s (cap 60 s); Whisper still pads Mel to 30 s.

### C. Readout

Apply re-runs the decoder. Positions come from the model’s generation routine (Whisper’s decoding mixin) or from a provided transcript. A teacher-forced pass collects  $h_{\ell,t}$ . Each cell is

$$\text{lens}(h_{\ell,t}) = \text{softmax}(E J_\ell h_{\ell,t}), \quad (3)$$

implemented as  $h J_\ell^\top E^\top$ . Whisper’s final LayerNorm is already inside  $h_{\text{final}}$ , so it is absorbed into  $J_\ell$ ; we do not apply an extra RMS-norm (unlike [2]). Top- $k$  (default  $k = 5$ ) ranks logits; the softmax is for display, not calibrated confidence. Neighboring cells are not a phrase.

The readout axis is transcript position, not audio time. Which encoder frame supports a token remains a cross-attention question.

### D. Why encoder transports fail

An earlier design fitted a global map from the mean-pooled encoder state to the mean-pooled decoder state and applied it to 0.3 s encoder buckets (Table I). Differentiating after pooling erases  $(t, t')$  alignment; the linearization is around clip means but is applied locally; the unembedding is used off-manifold with the constant dropped. That object is not (1). CTC support came for free (the encoder is the only stream) and disappeared with the redesign: a CTC head never verbalizes through a decoder workspace.

## IV. IMPLEMENTATION AND VERIFICATION

ECHO stores a format-v2 artifact: architecture *decoder*, method Hutchinson-decoder-VJP, one square  $d \times d$  matrix per recorded decoder block, bound to a model revision. Apply refuses a mismatch. Only seq2seq ASR adapters expose the architecture; wav2vec 2.0 returns none. Fig. 1 sketches the

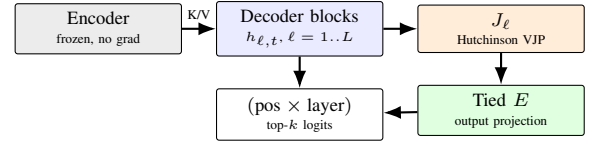


Fig. 1. Decoder-only J-lens in ECHO. The encoder supplies cross-attention keys and values only. Each decoder block has its own  $J_\ell$ ; readout uses the model’s tied  $E$ .

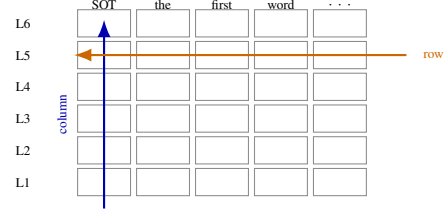


Fig. 2. Schematic (position  $\times$  layer) grid. A *column* is a depth trajectory at one token; a *row* is one block along the transcript. Cells are ranked verbalizable tokens, not a generated phrase.

data flow. A production fit on this codebase uses Whisper-base,  $N = 1000$  transcripts,  $P = 4$ , and writes six  $512 \times 512$  maps.

a) *Linear closed form.*: A unit-test stub uses a position-wise linear decoder  $h_{\text{final},t} = M h_{\ell,t}$  with no mixing across positions. Then  $\partial h_{\text{final},t'} / \partial h_{\ell,t}$  is  $M$  on the diagonal  $t' = t$  and zero otherwise, so the triangular mean in (1) is exactly  $2M/(T+1)$ . With  $P = 512$  Rademacher probes the fitted map matches this matrix to relative Frobenius error below 0.1. Apply on the same stub matches an independent recomputation of  $(Jh)E^\top$  at a held-out cell.

These tests check the estimator and the readout identity. They are not a Whisper workspace census.

## V. READING THE GRID

Fig. 2 is the contract of one apply job: rows are decoder blocks (UI “Layer  $\ell+1$ ” for 0-based  $\ell$ ), columns are decoder input tokens, each cell is a top- $k$  ranking. The following readings are *affordances of that contract*, not measured effects. We state them so the tool can be used without over-interpreting a cell.

**Start-of-transcript.** The first decoder input is the start token. No transcribed word is in the prefix. The residual still contains that token’s embedding and position; audio enters only through cross-attention. The cell is the closest available reading of what the audio wrote into the decoder before any transcript language model context.

**Depth trajectory.** A column can be read from early blocks (often frequent-token geometry) toward later blocks (closer to the emitted word). Where a content word first enters the top- $k$  is a candidate crystallization depth. A column that never ranks the emitted word is a candidate for transcription against the decoder’s internal ranking.

**Prior versus evidence.** Agreement of mid and late blocks on a collocation is consistent with a language-model prior. A late-block flip to an acoustically specific word is consistent

with audio-driven selection. This is a comparison of rankings, not a causal mediation test.

**Ambiguity, loops, endpoints.** A wide top- $k$  (their/there/they’re) is consistent with under-determined acoustics. Rankings of a repetition token before generation emits it would be a leading indicator of Whisper’s silence loop, if observed. Drift toward end-of-text or trained closers while speech continues would be an early-cutoff signature, if observed.

A layer-wise top-1 agreement curve—the speech analogue of the LLM workspace band [2]—is not reported here.

## VI. LIMITATIONS

$J_\ell$  is a first-order average: trust rankings more than scores, and expect error far from the fit distribution. Averaging over pairs erases which audio frame a token reads. Fit is teacher-forced, so  $J$  reflects “given this prefix,” not free-running dynamics.  $P = 4$  leaves probe noise; the closed-form test used  $P = 512$ . Frequent tokens dominate geometry. The artifact is revision-bound. Write-mode interventions from [2] are unimplemented. Layer count follows whatever HuggingFace records as decoder hidden states; it is six on current Whisper-base, not an embedding-plus-blocks stack.

## VII. CONCLUSION

The Jacobian lens is a statement about a residual that is about to be unembedded. In encoder–decoder ASR that residual is the decoder’s. Fitted there, with causal averaging and the model’s own  $E$ , it is the same estimator as in [2] and is already a job in ECHO. Fitted on pooled encoder states, it is not a lens. The measurements that would make the speech case as strong as the LLM case—workspace-band statistics, multi-size Whisper, and a write mode—should be taken on this decoder construction.

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